

Finite-scale memory and counting fluctuations in signed market order flow

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We give a finite-scale account of how temporal organisation amplifies the variance of signed market order flow. The presentation is designed to make the translation between market microstructure and nonequilibrium statistical physics explicit. Four years of hourly raw signed volume and volume-normalised imbalance are analysed in four primary cryptocurrency assets, with four additional assets kept as a post-hoc extension. The central result is an exact finite-sample decomposition of weekly variance into an analytic marginal reference, sign ordering at calendar-homogenised magnitudes, and a sign-magnitude remainder. Under the primary quarterly $\times$ hour-of-week conditioning, raw variance is amplified by 2.18–3.32 and the coupled remainder accounts for 65–79% of the excess; paired eight-week moving-block intervals remain above one half in all four primary assets. The attribution is scale-conditioned: monthly strata lower the coupled share to 45–49%, and two-point biweekly strata make it unidentifiable. Global variance growth remains superdiffusive under drift-resistant estimators, $\nu = 1.10$ –1.36, while a single-power correlation model fails and the local crossover is unresolved. We also identify an apparent post-event ageing signal that is reproduced by randomly placed events and caused by pre-event normalisation. Exact second-order and Gaussian identities are separated from measurements throughout, so neither accumulated variance nor a linear fluctuation asymmetry is counted as independent evidence.

I. INTRODUCTION

Market orders have a direction and a volume. A buyer-initiated trade removes liquidity from the ask; a seller-initiated trade removes liquidity from the bid. Summing their signed volumes over a time window produces a net flow. In the language of statistical physics this is a current, and the sum over a finite window is a transferred quantity. The vocabulary is useful because it brings response functions, noise, counting statistics, and finite-time nulls into one calculation. It does not turn a market into an electronic conductor or convert signed volume into entropy production.

Order-flow persistence is well established [1–3]. Order splitting explains how a large parent order can generate a long same-sign sequence, and account-level data support this mechanism quantitatively [4–6]. Recent work treats signed flow together with fluctuating volume, volatility, and price impact [7, 8]. Our question is narrower: at a specified horizon and calendar conditioning, what fraction of the excess variance is carried by sign ordering at a typical flow magnitude, and what fraction requires signs and magnitudes to retain their joint temporal organisation?

That distinction cannot be obtained by comparing two marginal autocorrelations. If $x_t = s_t m_t$, every off-diagonal contribution to the variance contains $s_t s_{t'} m_t m_{t'}$. Nor can two asymmetric surrogate tests be read as additive components. We instead construct a finite-sample identity with an analytic reference and an explicit interaction remainder. The result is the main empirical contribution of this article.

Three supporting results delimit its interpretation. First, accumulated-flow variance and the flow autocorrelation are exactly the same second-order information

written in two forms. Second, a linear $\log[P(Q)/P(-Q)]$ is automatic for a Gaussian variable and therefore is not, by itself, a fluctuation theorem. Third, local scaling and post-event clocks are sensitive to mean drift and normalisation. We use drift-resistant estimators, moving-block uncertainty, and random-event placebos to separate what is measured from what is an identity or an unresolved diagnostic.

Table I gives the translation used below. It is included for two readers at once: a finance reader need not know mesoscopic transport, and a physicist need not know how taker flow is recorded in a candle.

II. DATA AND OBSERVABLES

We use public Binance candles for BTC, ETH, BNB, and SOL against USDT over four years. Each hourly candle records total base volume V_t , taker-buy base volume V_t^+ , and trade count. We analyse

$$\varepsilon_t^{(r)} = 2V_t^+ - V_t, \quad \varepsilon_t^{(n)} = \frac{\varepsilon_t^{(r)}}{V_t}, \quad (1)$$

with $\varepsilon_t^{(n)} = 0$ when $V_t = 0$. The first is raw signed base volume; the second is dimensionless imbalance in $[-1, 1]$. A candle aggregates many trades, so $|\varepsilon_t|$ below is a magnitude of hourly residual flow, not the size of an individually identified market order.

The complete data contain 35 063 hourly observations per asset. The weekly analysis retains 206 complete Monday–Sunday UTC weeks, or 34 608 candles. XRP, ADA, DOGE, and AVAX form a post-hoc extension selected after the primary analysis was fixed. They broaden the empirical panel but do not enter the original four-asset decision rule.

TABLE I. Operational dictionary between market microstructure and transport. The final column states the limit of each translation.

Market quantity	Physics term	Measured meaning and limitation
Signed taker flow ε_t	Instantaneous current	Buyer-initiated minus seller-initiated executed volume in one candle. It is an aggregate observable, not electric charge or entropy production.
$Q_T = \sum_{t=1}^T \varepsilon_t$	Transferred quantity	Net signed flow accumulated over a declared window. No conservation law for a closed system is assumed.
$C_\varepsilon(\tau)$	Current-noise kernel	Temporal covariance of signed flow. Its double sum fixes $\text{Var}(Q_T)$ exactly.
$R(\tau)$	Retarded response diagnostic	Covariance between later price change and present flow. It is not identified causal metaorder impact or electrical conductance.
$\text{Var}(Q_T)/V_0$	Noise amplification	Variance relative to an analytic calendar-conditioned marginal reference. It is independent of an arbitrary transfer-size quantum.
$\log[P(Q)/P(-Q)]$	Fluctuation asymmetry	Finite-window distributional asymmetry. A linear form is automatic for a Gaussian and need not be thermodynamic.
Sign-magnitude remainder	Interaction contribution	Exact remainder after homogenising magnitudes in the off-diagonal sign term. It is a declared attribution, not a unique causal allocation.

We checked the candle construction against public individual-trade archives for one complete UTC day: 9,797,860 BTCUSD and 4,969,357 ETHUSD operations. Aggregation reproduces hourly volume, taker-buy volume, and trade count to floating-point precision. The individual trades also show why an absolute Fano level is not central here. The median individual sizes are smaller than the median hourly mean-size proxies by factors 87.5 and 71.6. Moreover, a compound-Poisson process already has a Fano factor proportional to $\langle q^2 \rangle / \langle q \rangle^2$, which can be large without memory. Ratios and the analytic reference introduced below avoid this arbitrary quantum.

Public candles do not reveal parent metaorders or trader identities. Results about order splitting are therefore mechanistic interpretations informed by the literature, not direct account-level measurements. This distinction is especially important because impact inferred from public flow can misassign the source of order-flow autocorrelation [9]. Our central variance calculation concerns the public flow series itself and does not reconstruct metaorder impact.

All windows are trailing, flow features are aligned to the candle in which they are observed, and response is referenced to the opening rather than the closing price. Perturbing future observations leaves past features unchanged; these alignment and no-look-ahead properties are tested in the public code [10].

III. MEMORY AND AGGREGATE RESPONSE

We use one sign convention throughout:

$$C_\varepsilon(\tau) \sim \tau^{-a}, \quad a > 0. \quad (2)$$

The raw hourly flow does not support one exponent over all resolved lags. Over $\tau \in [2, 10]$, the fitted fast ex-

ponents are $a_f = 1.00 \pm 0.15, 0.87 \pm 0.08, 0.46 \pm 0.05$, and 1.71 ± 0.23 for BTC, ETH, BNB, and SOL. Over $\tau \in [10, 50]$, the slow exponents are $a_s = 0.17 \pm 0.13, 0.22 \pm 0.09, 0.29 \pm 0.05$, and 0.16 ± 0.32 . Removing the daily seasonal mean changes neither regime materially [Fig. 1(b)]. The slow component is therefore not a non-decaying daily drive, although its exponent remains imprecise.

The lagged aggregate response

$$R(\tau) = \frac{\text{cov}(p_{t+\tau} - p_t, \varepsilon_t)}{\text{Var}(\varepsilon_t)} \quad (3)$$

is much flatter than C_ε [Fig. 1(a)]. This is consistent with the standard “surprise” mechanism: prices respond primarily to the innovation in persistent flow rather than to its predictable component [8, 11–13]. Some series have a small but resolved response slope, so “flat” is an approximation and not a universal law.

An earlier derivative ratio $-C'_\varepsilon/R$ was too noisy to support an empirical fluctuation–dissipation claim and has been removed from the result set. Differentiating a sample correlation and dividing by an estimated response requires joint uncertainty that the data do not resolve. Nothing below uses that ratio.

IV. COUNTING FLUCTUATIONS AND EXACT SAFEGUARDS

For a window of T candles define

$$Q_T = \sum_{t=1}^T \varepsilon_t, \quad s_T(Q) = \log \frac{P(Q_T = Q)}{P(Q_T = -Q)}. \quad (4)$$

The notation comes from full counting statistics [14–17]; here it simply means the distribution of accumulated

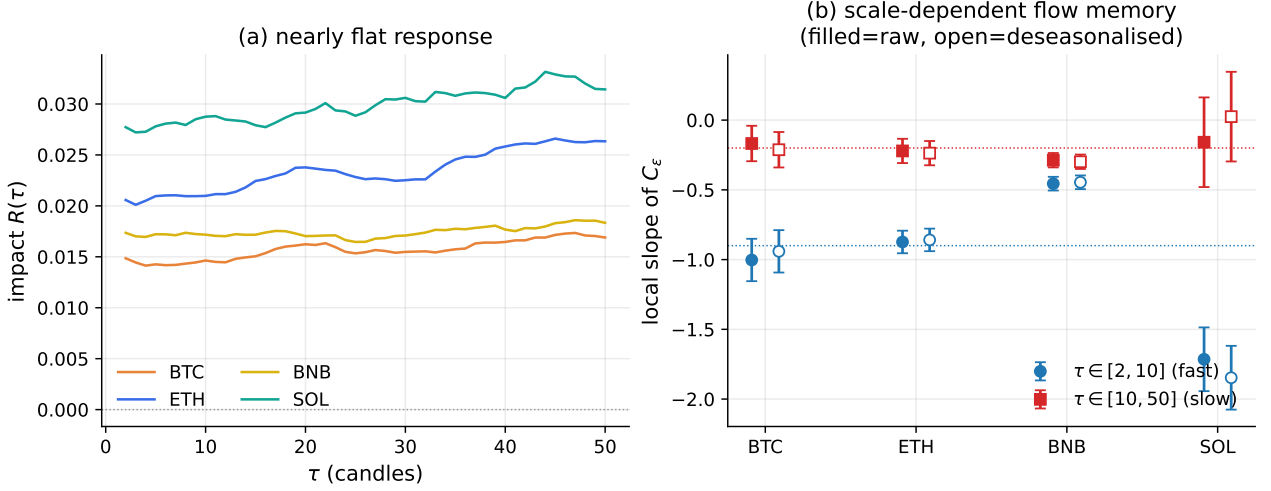


FIG. 1. Memory and aggregate response at hourly resolution. (a) The lagged response varies slowly compared with the flow correlation. (b) Local correlation exponents in fast and slow lag windows, before and after removing the daily seasonal mean. We use $C_\varepsilon \sim \tau^{-a}$ with $a > 0$ throughout.

market flow.

Two identities prevent double counting of evidence. First,

$$\text{Var}(Q_T) = \sum_{t,t'=1}^T C_\varepsilon(t-t') = TC_\varepsilon(0) + 2 \sum_{\tau=1}^{T-1} (T-\tau)C_\varepsilon(\tau). \quad (5)$$

Thus variance growth and the autocorrelation are two representations of one second-order object, not independent confirmations.

Second, if Q_T is Gaussian with mean m_T and variance V_T , then

$$s_T(Q) = \frac{(Q + m_T)^2 - (Q - m_T)^2}{2V_T} = \frac{2m_T}{V_T}Q \quad (6)$$

exactly. A measured linear fluctuation asymmetry with slope $2\langle Q_T \rangle / \text{Var}(Q_T)$ therefore has a Gaussian algebraic explanation. It is not sufficient evidence for a Gallavotti–Cohen theorem [18, 19]. The data are not uniformly Gaussian: standardised skewness and excess kurtosis do not converge to zero across all assets and scales. We consequently use Eq. (6) as a null explanation, not as a fitted distributional model.

There is also no established current large-deviation principle. The usual speed is T , whereas the variance here grows as T^ν with $\nu > 1$. We do not infer entropy production, thermodynamic affinity, or a market temperature from $s_T(Q)$.

V. EXACT SIGN–MAGNITUDE ATTRIBUTION

This section contains the central measurement. Complete weeks are demeaned within a declared period $\times$

UTC hour-of-week stratum $g(t)$, giving $x_t = s_t m_t$. Let

$$Q_w = \sum_{t \in w} x_t, \quad V_{\text{obs}} = \text{Var}_w(Q_w). \quad (7)$$

Every complete week contains at most one observation from each stratum. Under independent within-stratum permutations, all expected off-diagonal weekly products vanish, while the diagonal sum is preserved. The exact expectation of the joint-order null is therefore

$$V_0 = \frac{1}{W-1} \sum_t x_t^2. \quad (8)$$

This replaces a Monte Carlo denominator by a finite-sample identity.

To isolate temporal sign organisation on the same calendar footing, center the signs and homogenise only the magnitudes in the off-diagonal term:

$$\tilde{s}_t = s_t - \langle s \rangle_{g(t)}, \quad \bar{m}_{g(t)} = \langle m \rangle_{g(t)}, \quad (9)$$

$$C_s = \sum_w \sum_{\substack{t,t' \in w \\ t \neq t'}} \tilde{s}_t \tilde{s}_{t'} \bar{m}_{g(t)} \bar{m}_{g(t')},$$

$$V_s = \frac{C_s}{W-1}, \quad V_{sm} = V_{\text{obs}} - V_0 - V_s. \quad (10)$$

Equation (10) reconstructs V_{obs} exactly. The fractions

$$f_s = \frac{V_s}{V_{\text{obs}} - V_0}, \quad f_{sm} = 1 - f_s \quad (11)$$

are reported without clipping. The interaction allocation is conditional on sign centering and stratum-wise magnitude homogenisation. The observed diagonal remains in

V_0 , so $V_0 + V_s$ is not the ordinary variance of the homogenised proxy. Other declared stratum scales would produce other exact remainders. We therefore present a fixed-diagonal operational attribution, not an orthogonal or uniquely causal decomposition of a market mechanism.

With quarterly $\times$ hour-of-week conditioning, raw-flow amplification is 2.75, 3.32, 2.18, and 2.53 for BTC, ETH, BNB, and SOL [Fig. 2(a)]. The sign shares are 0.21, 0.30, 0.35, and 0.35; the coupled shares are 0.79, 0.70, 0.65, and 0.66. Paired eight-week moving-block intervals for f_{sm} are [0.63, 0.88], [0.62, 0.78], [0.52, 0.76], and [0.55, 0.74]. The coupled remainder is therefore a majority in all four primary raw point estimates and throughout their intervals.

Normalised imbalance has $f_s = 0.41$ –0.51 and $f_{sm} = 0.49$ –0.59. Normalisation raises the sign-share point estimate in every primary asset, but paired intervals resolve the increase clearly only for BTC. This is a directional consistency observation rather than an independent universal test.

The attribution is scale-conditioned. Monthly strata lower the primary raw coupling share to 0.45–0.49. A 14-day partition leaves at most two observations per hour-of-week stratum; after demeaning, their residual magnitudes are equal and V_{sm} is forced to zero. That case is unidentifiable, not evidence that coupling vanishes. The quarterly–monthly contrast places much of the resolved interaction in multiweek organisation absorbed by finer conditioning.

The post-hoc raw coupling shares are 0.52–0.73, but two of four intervals cross one half. We use them to support a non-negligible interaction, not an eight-asset majority claim. Cross-asset weekly correlations also show that the series are not independent replications. Finally, the original predeclared gate required raw observed variance to exceed twice the 98.75th joint-null percentile in all four primary assets; BNB failed. The exact attribution answers a different question and does not rewrite that outcome.

Uncertainty uses 3999 paired circular moving-block replicates with block lengths 4, 8, 13, and 26 weeks. The same resampled week indices enter all components, and stratum estimates are held fixed. Full tables, deposited outputs, and reconstruction checks are available in the public repository [10].

VI. SCALING: ROBUST LEVEL, UNRESOLVED CROSSOVER

For a single kernel $C(\tau) \sim \tau^{-a}$ with $0 < a < 1$, Eq. (5) gives $\text{Var}(Q_T) \sim T^{2-a}$. At $a = 1$ the leading form is $T \log T$; for $a > 1$ the correlation is summable and the asymptotic variance is diffusive. The domain distinction matters: the uncertainties of the BTC and SOL fast exponents cross $a = 1$, so a simple $\nu = 2 - a$ band is not uniformly defined for them.

A broad-window single-power prediction misses all

TABLE II. Global variance exponent over $T = 8 \dots 256$. The wavelet columns remove constant, linear, and quadratic drift, respectively.

asset	block	db1	db2	db3
BTC	1.42	1.20	1.14	1.13
ETH	1.37	1.24	1.26	1.29
BNB	1.43	1.35	1.36	1.30
SOL	1.31	1.15	1.13	1.10

eight primary asset–resolution series [Fig. 3(a)]. Predicted exponents are 1.45–1.88, while block estimates over $T = 1 \dots 256$ are 1.22–1.37. Restricting the block fit to $T = 8 \dots 256$ gives 1.31–1.43; the two ranges refer to different grids and are not interchangeable. Two-regime point-estimate bands are only consistency checks because their amplitudes and crossover uncertainty were not fitted. BNB’s $\nu = 1.37$ lies outside its [1.54, 1.71] band [Fig. 3(b)].

The apparent local exponent rises with scale in the block estimator, but a drifting block mean contributes a T^2 term with the same qualitative signature. We therefore compare block variance with Daubechies filters having one, two, and three vanishing moments. Applied directly to the flow, db1 removes a constant mean, db2 a linear trend, and db3 a quadratic trend [20, 21]. All four estimators agree within 0.01 on exact fractional Gaussian noise of known ν ; injected drift moves only the estimator that should be sensitive.

On the data, drift-resistant estimates span 1.10–1.36 [Table II], below the block values in every asset. Thus global superdiffusion is robust, but the local crossover is not resolved. At $T = 256$ only 135 non-overlapping windows remain in the complete-week sample, making the longest scales the noisiest. BNB also turns over at the long end instead of continuing the proposed monotone drift. A longer record or a stable-mean asset would be needed to discriminate the crossover.

VII. POST-EVENT CLOCKS AND A NORMALISATION ARTEFACT

We define shocks as absolute returns above the 99.5th percentile and construct 240 hourly post-shock trajectories. The mean normalised absolute return is compatible with relaxation, but an Omori-amplitude fit gives $\beta = 0.61 \pm 1.01$ and six of ten asset–resolution fits terminate at an optimiser boundary. The uncertainty includes zero, so no universal relaxation exponent is identified [Fig. 4(a)].

The correlation time appears to grow monotonically with waiting time, with a Spearman coefficient of +1.00. Repeating the entire procedure at randomly placed event times returns the same median coefficient [Fig. 4(b)]. The apparent ageing is therefore produced by the measure-

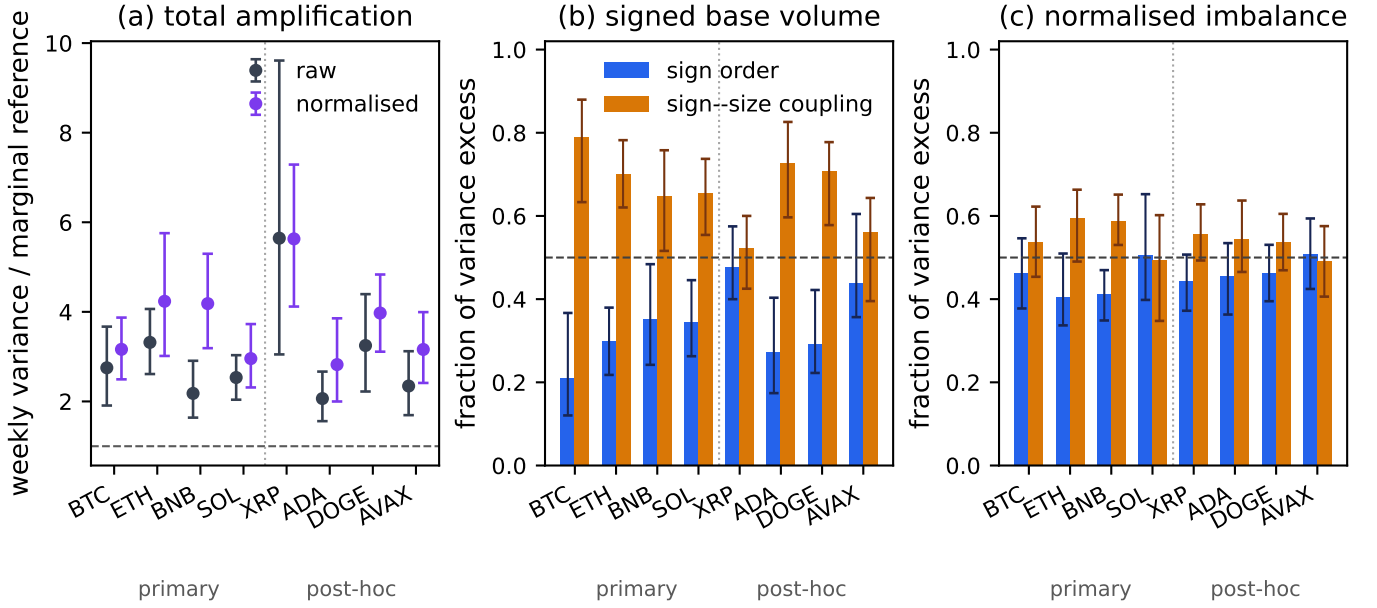


FIG. 2. Total weekly amplification and exact attribution under quarterly $\times$ hour-of-week conditioning. (a) Variance relative to the analytic marginal reference. (b),(c) Fractions of the excess due to sign order at homogenised magnitudes and to the sign-magnitude remainder. Error bars are paired eight-week moving-block 95% intervals. The first four assets are the primary panel; the remaining assets are post hoc.

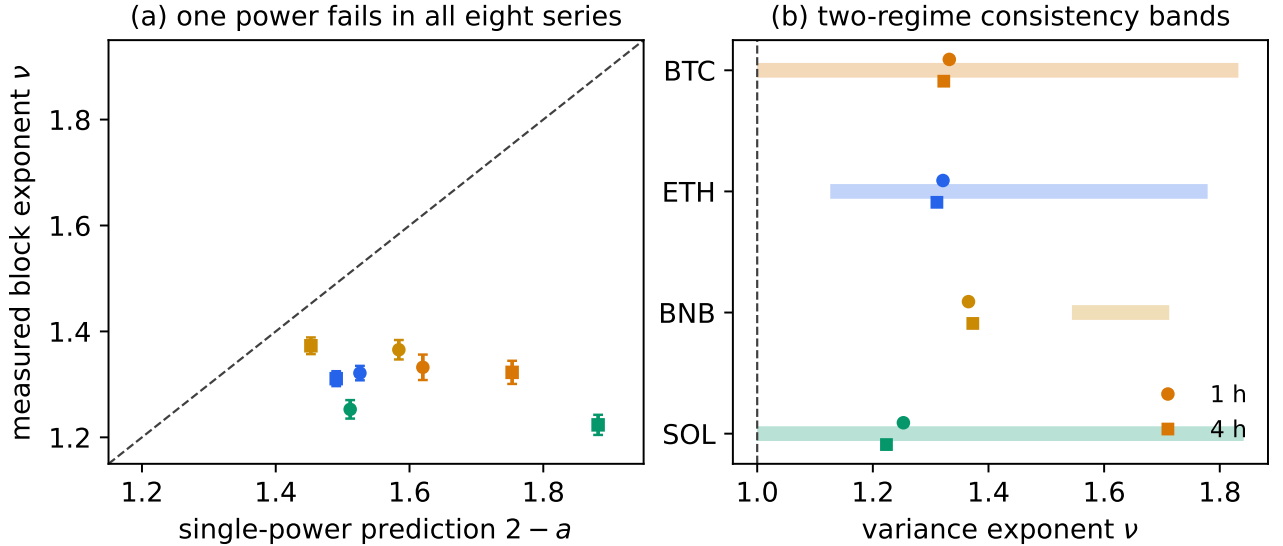


FIG. 3. Scaling reassessment. (a) The broad-window single-power prediction misses all eight hourly and four-hourly series. (b) Asset-specific two-regime consistency bands; BNB lies outside. The bands omit amplitudes and crossover uncertainty and are not quantitative predictions.

ment convention: dividing a post-event signal by a noisy pre-event scale injects a slow common mode as the process moves away from that reference.

This is a ratio-normalisation problem related to classical spurious-correlation bias [22–24] and to event-conditioned correlation distortions [25, 26]. In a granular cooling state, normalised two-time correlators similarly exhibit apparent ageing from self-similar scale decay even

without memory [27]. Random-event placebos are therefore necessary whenever waiting time and normalisation share the same event origin. Our result is limited to this construction; it does not invalidate ageing measured with a different estimator.

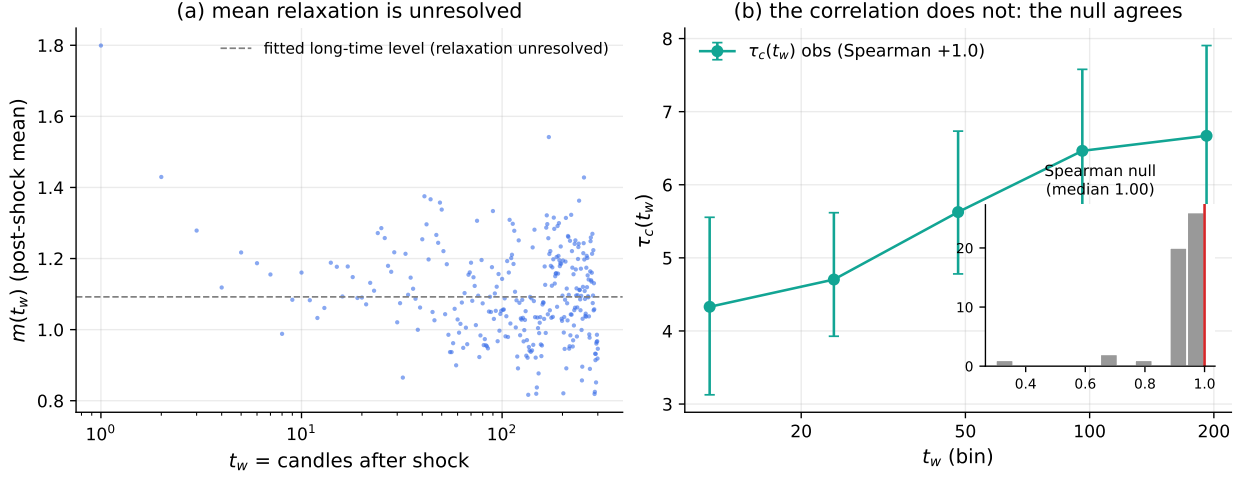


FIG. 4. Post-event clocks. (a) Mean relaxation is unresolved: the fitted amplitude exponent includes zero and many fits reach optimiser bounds. (b) Correlation time appears to increase with waiting time, but randomly placed events reproduce the same monotone statistic, identifying a normalisation artefact.

VIII. A COMPACT STOCHASTIC REPRESENTATION

The empirical results can be organised without a microscopic market field theory. Let the demeaned flow have a Wold representation [28, 29]

$$\varepsilon_t = \sum_{k \geq 0} h_k w_{t-k}, \quad \langle w_t w_{t'} \rangle = \sigma_w^2 \delta_{tt'}, \quad h_0 = 1, \quad (12)$$

where w_t is the innovation not linearly predictable from past flow. A minimal price equation

$$p_{t+1} - p_t = \lambda w_t + \eta_t \quad (13)$$

immediately gives a lag-independent covariance between future price change and present flow for positive lags. This is the familiar surprise model, stated here only to connect the nearly flat response to finance language. It is a benchmark consequence of the assumed innovation coupling, not new evidence.

If the flow were Gaussian, adding a counting field χ would give

$$F_T(\chi) = \log \langle e^{\chi Q_T} \rangle = \chi m_T + \frac{1}{2} \chi^2 V_T. \quad (14)$$

Equations (5) and (6) then follow immediately. This compact representation clarifies the logical status of the calculations: the response and Gaussian identities are closures under declared assumptions, whereas the sign-magnitude attribution and its intervals are measurements.

IX. INFERENCE RULES USED IN THE ANALYSIS

Three practices materially changed the conclusions.

Paired resampling. Ratios built from the same weeks are resampled with the same block indices. Marginal confidence intervals do not answer whether two dependent components differ.

Positive controls. A null result is accompanied by an injection-recovery test on synthetic data carrying the proposed signal. Otherwise “no effect” is indistinguishable from an insensitive pipeline in a null analysis.

Multiplicity at scanned scales. In an exploratory grid of 144 related configurations, the smallest nominal p value was 2×10^{-4} , yet the Westfall-Young maximum-statistic correction gave a global $p = 0.43$ [30]. Scale scans are therefore treated as model diagnostics, not as 144 independent opportunities for discovery.

All numerical claims in the text are generated from deposited JSON outputs; the code checks exact reconstruction, fixed seeds, cache hashes, and no look-ahead. The archived outputs also preserve failed predeclared gates and withdrawn exploratory analyses rather than replacing them with later choices [10].

X. DISCUSSION

The main result is not that signs dominate. At quarterly conditioning, sign ordering at homogenised magnitudes explains only 21–35% of the primary raw weekly excess. The remaining 65–79% requires the actual temporal organisation of signs with hourly flow magnitude. This is consistent with order splitting, in which a parent order can generate both a same-sign run and elevated executed flow, but the public candle data do not identify that microscopic mechanism directly.

The word “finite-scale” is essential. Monthly conditioning absorbs enough low-frequency organisation that the coupled share is no longer a resolved majority, and bi-

weekly conditioning destroys identifiability. The result is a property of a declared observable, horizon, venue, and conditioning scheme, not a universal constant of markets.

The same discipline applies to the other sectors. Global superdiffusion is robust under drift-resistant estimators, but the detailed crossover is open. The event-conditioned correlation does not establish ageing because

its placebo produces the same statistic. The accumulated variance is not independent of the autocorrelation, and a Gaussian linear fluctuation asymmetry is an identity. Stating these boundaries makes the positive result easier to see: the resolved empirical content is a large, scale-conditioned sign-magnitude contribution to finite-window order-flow noise.

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